

AWS checklist and maintenance guide

Station name	ASL TAS-L	
People present	MOLA, MCIT	instructed by:
Purpose	<u>standard maintenance</u> / installation / take-down / quick visit / emergency	
Date	3/8/2019	
Transport	<u>helicopter</u> / fixed wing / snow mobile / dog sleds / on foot from:	
Weather	<u>warm</u> / cold / very cold / <u>no wind</u> / some wind / rain / cloud cover %: /6	

Notes:

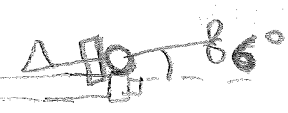
Arrived directly from TAS-A where we camped. Arrived 7.45

TAS-2

Metadata before maintenance (if establishing a new AWS: skip this page)

Time difference between logger clock and UTC	AWS LAB 30 S + / -
Adjust logger time to UTC (be sure PC runs on UTC time)	☒ Y / ☐ N
Download data to PC and/or change CF Card (wait for green light)	☒ Y / ☐ N
Name of logger program	2013C
Photo or screen dump of values in fast scan mode	☒ Y / ☐ N
Latitude (dd mm.mmm)	65° 38.433' N
Longitude (dd mm.mmm)	38° 53.904' W
Photos of tripod, sensors, logger box wiring etc. (more is better)	☒ Y / ☐ N
Mast tilt in boom direction (+ if radiometer tilting down)	8 °
Mast tilt across boom looking from radiometer (- if clockwise)	- 2.9 °
Boom direction relative to north	8 ° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.	SEE SKETCH ☒ Y / ☐ N
Radiometer aligned with mast and inclinometer? If not, measure.	☐ Y / ☐ N
Temperature / humidity height (bottom of casing – surface)	260 cm
Sonic ranger on AWS height (membrane – surface)	278 cm
Wind sensor height (center of propeller – surface)	383 cm
Sonic ranger on stakes height (membrane – surface)	345 cm
Free length of stake – outer, holding boom with sonic ranger	358 cm
Free length of stake – middle	341 cm
Free length of stake – outer, holding boom without sonic ranger	349 cm
Free length of ablation hose on/above the ice surface	16 m 20 cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	0 cm (empty)
Vertical difference in surface height (at station – at hose) (approx)	30 cm
Length of thermistor string on surface from surface marking	6 m 80 cm
Height of surface irregularities	~1.5 m cm
Stake assembly still standing?	☒ Y / ☐ N / more or less
Apparent damage to sensors? If yes, which?	☒ Y / ☐ N

Notes:

↓
 Press sensor hose empty/drain
 Wind monitor not aligned with mast: 
 alignment corrected during maintenance

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	080202	=	- Aligned with mast: ☒ Y/N - Clean: ☒ Y/N
Inclinometer	00634	=	- Aligned with radiometer within 0.5°: ☒ Y/N
Satellite antenna	60528	=	
Wind sensor	110445	=	- Box on sensor exactly towards mast (south): ☒ Y/N <i>fixed</i>
Temperature / humidity assembly			
T/hum assembly casing	=	MARKS: 61154942	- Fan spins and sounds OK: ☒ Y/N
T/hum assembly bracket	=	RS12T: 61138896	
Temperature probe in T/hum assembly	=	HP102H: 61154941	
HygroClip in T/hum assembly	H40	HCL53: 61180277	- 40 °C offset in HygroClip temperature: ☒ Y/N
Sonic ranger on AWS	UNREADABLE	=	- Old sensor with new membrane: ☒ Y/N - Clicking 5 times as it should: ☒ Y/N
Solar panel	UNREADABLE	=	- Output OK: ☒ Y/N - Clean: ☒ Y/N
GPS antenna	GENS 04	=	- Stuck to top of logger enclosure: ☒ Y/N
Data logger	E3036 21576	=	- New internal battery in old logger: ☒ Y/N - New operating system: ☒ Y/N
Card reader	4592	=	- LED green of flashing orange (active): ☒ Y/N
Multiplexer	E3334	=	- Clicking as it should: ☒ Y/N
Iridium modem IMEI number	16275P0	16275P	- Transmissions tested: ☒ Y/N - Problems: ☒ Y/N
Barometer	3215015	=	
Sonic ranger on stakes	UNREADABLE	5960	- Old sensor with new membrane: ☒ Y/N - Clicking 5 times as it should: ☒ Y/N - In danger of being buried next winter: ☒ Y/N
Thermistor string	B08	B20	- Old string left in ice: ☒ Y/N/cut
Ablation hose assembly	5018ABS	5026ABS	- Old hose left in ice: ☒ Y/N/cut - Air in hose: ☒ Y/N/removed <i>NEW OK</i> - Bladder 50% full: ☒ Y/N/ after emptying/filling <i>NEW OK</i>
Logger enclosure	- Replaced (including everything inside): ☒ Y/N - New vent filter in logger enclosure: ☒ Y/N - Moisture inside: ☒ Y/N - New desiccant bags (2): ☒ Y/N		
Battery box	- New box with new batteries: ☒ Y/N - Old box with new batteries: ☒ Y/N - Moisture inside: ☒ Y/N <i>UNOPENED</i> - New desiccant bags (2): ☒ Y/N		
Stakes	- New ones drilled in: ☒ Y/N - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: ☒ Y/N		
Tripod	- New parts: ☒ Y/N - Boom direction: true north-south: ☒ Y/N		

Metadata after maintenance / establishment

Time logger clock set (to UTC)?	☒ Y ☐ N
Name of logger program	- 2014d
Photo or screen dump of values in fast scan mode	☒ Y ☐ N
Latitude (dd mm.mmm)	SAME
Longitude (dd mm.mmm)	SAME
Photos of tripod, sensors, logger box wiring etc. (more is better)	☒ Y ☐ N
Mast tilt in boom direction (+ if radiometer tilting down)	+ 4.6 °
Mast tilt across boom looking from radiometer (- if clockwise)	- 0.5 °
Boom direction relative to north (radiometer should point to true south)	13 ° true/magn
Box on wind sensor exactly towards mast? If not, change.	☒ Y ☐ N
Radiometer aligned with mast and inclinometer? If not, align.	☒ Y ☐ N
Temperature / humidity height (bottom of casing – surface)	265 cm
Sonic ranger on AWS height (membrane – surface)	274 cm
Wind sensor height (center of propeller – surface)	317 cm
Sonic ranger on stakes height (membrane – surface)	87 cm
Free length of stake – outer, holding boom with sonic ranger	116 cm
Free length of stake – middle	120 cm
Free length of stake – outer, holding boom without sonic ranger	127 cm
Free length of ablation hose on/above the ice surface	14 m 75 cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	136 cm
Vertical difference in surface height (at station – at hose) (approx)	AWS + 20 cm
Length of thermistor string on surface from surface marking	46 cm
Height of surface irregularities	SAME cm

Notes:

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